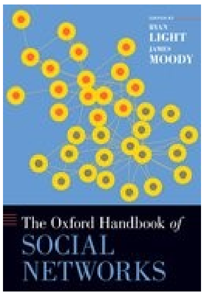

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CHAPTER

10 The Continued Relevance of Ego Network Data

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Abstract

Ego network data have a long history in the social sciences, acting as a bridge between traditional statistical techniques and network analysis. Ego network data provide personal network information, as the data are based on a sample of individuals. This chapter details the basic features of such data, describing the advantages, disadvantages, and potential applications of using ego network data. Ego network data remain a popular choice, despite the growing availability of full network data sources. This is in large part because ego network data are easy to collect but still provide a surprisingly large amount of network information. Ego network data are also quite flexible, with past work using the same basic data structure for widely different purposes. Given the ease of collection and the flexibility of use, there is every reason to believe that ego network data will continue to be a useful option for network scholars.

Keywords: ego networks, sampling, homophily, social support, simulation
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p. 170 EGO network data have a long history in the social sciences, acting as a kind of bridge between traditional statistical techniques and network analysis (Perry, Pescosolido & Borgatti 2018). Ego network data continue to be collected and analyzed and are widely used in studies of health, organizations, and stratification (Cornwell, Laumann, & Shumm, 2008; Kadushin, 2012; Smith, McPherson, & Smith-Lovin, 2014). Researchers continue to use ego network data because (1) the data are (relatively) easy to collect and (2) the data provide a surprisingly large amount of relational, or network, information "on the cheap" (Smith, 2012). Ego network data are also quite flexible, with potential applications varying greatly in scale (individual to full networks) and type. For example, ego network data are often used to measure the social support available to individuals (Fischer, 1982; Wills & Shinar, 2000; Cornwell et al., 2009); the same basic data structure has, however, also been used as a purely methodological tool, used to correct the estimates of nontraditional sampling schemes (Lu, 2013). With its ease of collection, depth of information, and flexibility in use, the view here is that ego network data are, and will continue to be, useful for network scholars, even in the face of new and more widely available complete network data sources, such as from cell phone databases and online platforms.

Ego network data are generally defined in contrast to full network data (Marsden, 2011). With full network data, a researcher has information on all actors in a setting: one can identify each actor and determine whether a tie exists between each pair of actors. Under such ideal conditions, it is straightforward to calculate global measures of interest, such as group divisions and cohesion, as well as individual measures, like centrality (an individual's position in the network) (Freeman, 1979; Moody & White, 2003). Often, however, it is impractical to collect full network data on the population of interest (Frank, 1971; Smith, 2015). The network may be too large and the resources available too scant to interview everyone in the population, and an appropriate electronic data source may not exist. A researcher may be interested in the entire US population, for example, where collecting a full census on a particular relationship of interest (which may not exist in online sources) is not
p. 171 generally ↳ possible. It would also be difficult (although not impossible) to collect full network data across many different contexts, such as schools or neighborhoods.

Ego network data offer an alternative, basing the network information on a sample of individuals (e.g., McPherson, Smith-Lovin, & Brashears, 2006). The survey randomly samples individuals from a known population (generally assumed to have a known sampling frame), gathering information about the respondents and their local social network (Marsden, 1990). The data yield personal networks, where the focus is on ego, the focal node/respondent, and ego's immediate social neighbors, or alters (i.e., the people that ego is connected to). For example, a researcher may ask respondents whom they talk with on a regular basis, collecting data on each named alter; we would know the number of alters as well as the characteristics of each alter. The data will also often include the social connections between alters (although not always).

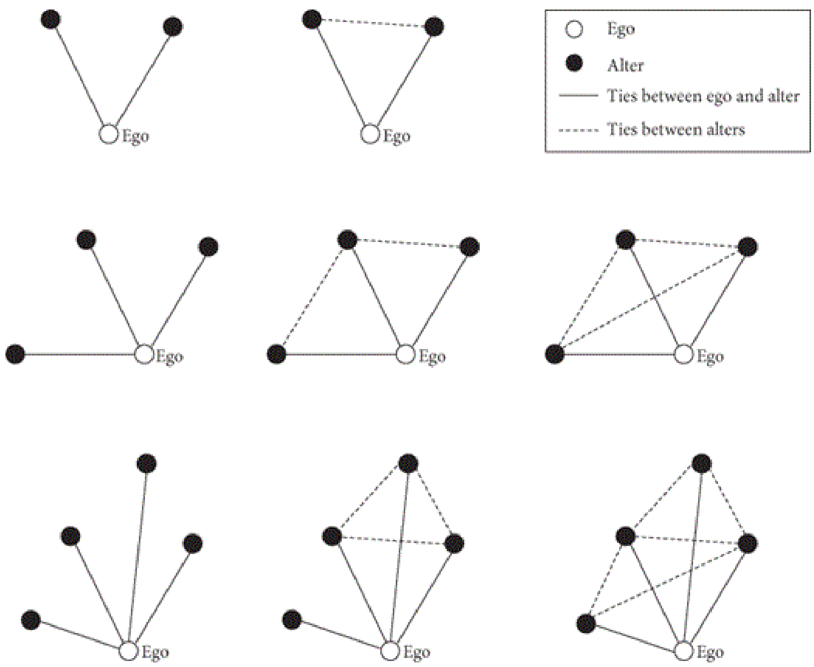

Figure 10.1 Example ego networks.

Figure 10.1 depicts a set of example ego networks. Ego represents the sampled respondent, from which the rest of the network information is derived. The dark black lines represent ties between ego and alter, while the dotted lines represent ties between alters. We can see in the top left corner that ego has two alters, who are not tied to each other. Or, in row 2, the far right ego has three alters where every alter is connected to every other alter. It is important to remember that the ego networks are sampled separately, with reports on immediate

contacts only. There is no information on the alters of the named alters. Similarly, ego network surveys do not
p. 172 collect identifying information on the named alters and there ↳ is no (simple) way of knowing if the sample respondents themselves are connected. Thus, the sampled parts of the network (collected within an ego network survey) cannot be connected. In that sense, one can think of ego network sampling as capturing pieces of the whole network, with each ego network representing a different piece of the puzzle (Smith, 2012; Smith et al., 2014).

Advantages and Disadvantages of Ego Network Data

There are many positive characteristics associated with personal network data, as well as a number of potential drawbacks. I begin with the advantages of ego network data before turning to the disadvantages.

Advantages

The data are, first and foremost, easy to collect. Rather than require information on every node and every tie between nodes, ego network data are only based on an individual's social network. This means that we do not need a census of all cases but can, instead, simply collect a sample of respondents. One need not have data on all Americans to collect useful, representative network data on the country (e.g., Marsden, 1987). Moreover, the ego networks are treated as independent, with no identifying information needed to link the ego networks. This greatly reduces the collection burden of the researcher.

Second, ego network data are easy to embed in traditional surveys. Ego network data are based on a random sample of the population. This makes it easy to add to an existing, typical survey. A researcher would still begin by taking a random sample of individuals and asking questions about demographics, socioeconomic status (SES), health, and the like. The only difference here is that one adds network questions, eliciting information about the respondents' social contacts. In fact, national surveys, like the General Social Survey (GSS), will periodically include ego network sections within the larger questionnaire (Burt, 1984). This embedding of ego network questions within larger surveys means that network data can be collected alongside (and thus used as a predictor of) measures of health, income, and the like.

Third, ego network data are easy to combine with alternative data collection strategies. For example, it is possible (and often very useful) to pair respondent-driven sampling (RDS) data with ego network surveys (Lu, 2013). With RDS, individuals from hidden populations (such as female sex workers) recruit others into the study, thus creating a chain of referrals that serve as the base sample (Salganik & Heckathorn, 2004). Additional ego network data can be collected on the recruited individuals, supplementing the RDS chain data in important ways (Lu, 2013; Verdery et al., 2015). Ego network data can also supplement more automated data sources like cell phone data, colocation Bluetooth data, or online data, fleshing out the local network characteristics of the sampled respondents (Golder & Macy, 2011; Oliver, Matic, & Frias-Martinez, 2015).

p. 173 Fourth, ego network data can be collected on sensitive relations, where it is difficult or impossible to have respondents identify their alters on the relation of interest. For example, it may be difficult to ask respondents to name their sexual partners, making it hard to collect complete networks (Krivitsky & Morris, 2015). Respondents may be reluctant, while no list (or roster) may exist for them to identify the partner. Similarly, given the confidential nature of the data, such collection efforts (on a third party, the named alter) may be met with institutional review board (IRB) complications. An ego network survey offers a nice alternative, as it does not require identifying the names of partners, only the number of partners and their characteristics. The researcher can thus collect network information without broaching difficult confidentiality issues, making it easier to elicit network information on sensitive relations like sexual contact and drug use (Morris & Kretzschma, 2000; Merli, Moody, Mendelsohn, & Gauthier, 2015).

Fifth, ego network surveys contain a great deal of information, including the characteristics of ego, the characteristics of the alters (or the people ego interacts with), the structural features of the personal network (e.g., size), and even something about the nature of the relationship between ego and each alter (closeness, frequency of contact, etc.). More subtly, ego network data offer information at both the micro and macro levels. At the micro level, there is information about individual respondents and the people they interact with. This local information can then be aggregated to infer characteristics about the population-level network structure from which the respondents were sampled, thus offering information about the macro level (Smith, 2012; Gjoka, Smith, & Butts, 2014).

Disadvantages

Such advantageous characteristics come at a cost, however. First, the data are based (typically) on self-reports. Ego (the focal respondent) reports on the characteristics of the alters and the ties between alters. There could be biases in such reporting, as respondents may be unable to report on their alters with complete accuracy; while demographic characteristics like gender and education are unlikely to cause many problems, respondents may, for example, be unable to accurately report on the political attitudes of their alters (Marsden, 1990). There may also be cognitive biases in the reporting of the alter-alter ties; for example, respondents may assume their alters know each other based on a cognitive desire to minimize conflict or divides within their ego network (Krackhardt & Kilduff, 1999).

Second, respondents must be treated as independent. Ego network data capture purely personal networks, based on an independent sample of respondents. The egos themselves are thus assumed to be independent and, in fact, must be treated so as there is (generally) no way to tell which egos are tied to each other. Such assumptions work well in nationally representative surveys, where it is extremely unlikely that any two selected respondents know each other (Smith et al., 2014). This works less well in smaller, bounded settings where a large number of respondents within a school or organization are sampled. If sampled egos are, in fact, tied together, then models that assume they are independent may have biased estimates and deflated standard errors (Ebring & Young, 1979).

Third, the data do a rather poor job of capturing asymmetries in social relations, as the relations are based only on ego's report of a tie. Many social connections must assumed to be symmetric (such as friendship), even if an
p. 174
underlying asymmetry actually exists (where ↳ A likes B much more than B likes A) (Gould, 2002; Smith & Faris, 2015). Additional data would have to be collected or the relation of interest must be concrete enough to allow for clear asymmetries in the interactions (e.g., Did you borrow money from each alter? Did they borrow money from you?).

What Can Be Extracted from Ego Network Data?

Table 10.1 summarizes the information available from ego network data. While one must weigh the potential drawbacks of using sampled data, the upside, in terms of information collected, is higher than one might expect. The first column in Table 10.1 describes the information collected from the survey itself. The second column lists the measures associated with each piece of information, while the third column offers example applications.

Table 10.1 Information, Measures, and Applications Associated with Ego Network Data

Information Collected	Example Measures	Example Substantive Applications
1. List of alters	Degree	Social support; risk factors
	Degree distribution	Global network inference
	Differential degree	Global network inference
2. Alter characteristics	Proportion same as ego	Social boundaries
	Mixing matrix	Social boundaries; global network inference
	Distributional summaries (e.g., proportion female)	Adjust RDS estimates; social support; risk factors
3. Tie characteristics	Distributional summaries (e.g., proportion kin)	Social support
4. Alter-alter ties	Density	Social support; risk factors
	Efficiency	Information flow and brokerage
	Ego network configurations	Global network inference

Respondents are first asked to list a set of social contacts, the alters. Practically, studies will often ask respondents to list a limited number of alters (e.g., up to 10), but it is generally best not to truncate the data in any way (Smith, 2015). Similarly, it is generally a good idea to ask more than one question to elicit the initial list of alters (Marin & Hampton, 2007). The person you hang out with may be different than the person you discuss health matters with. Multiple relations capture a fuller picture of the local personal network. The exact relations depend on the question of interest. Questions are typically either general (Looking back over the last six months, who are the people with whom you discussed an important personal matter?) or behavioral (Who have you slept with in the last six months?) or capture support relations (If you were sick, who would be willing to accompany you to the hospital?).

p. 175 Such data, at the individual level, offer an estimate of the number of contacts for ego, or individual degree. More generally, the listing of alters (aggregated over all respondents) offers an estimate of the degree distribution, equal to the number of alters per respondent. The alter list also offers information on differential degree, or the mean degree across demographic groups. This is inferable as there is information on degree and the demographic characteristics of the respondents. For example, those with higher levels of education may have more social ties than those with less education (e.g., Lizardo, 2006; McPherson et al., 2006).

The second row in Table 10.1 summarizes the information surrounding the alter characteristics. A researcher may ask the respondent to report on the race, gender, and age of the named alters (often only a subset of the total list). This information can be used in a number of ways. For example, we can ask if the respondents are similar to their social contacts. The respondent provides information on both their characteristics and their alters' characteristics; this makes it possible to ask if ego is the same/different than the alters (e.g., what proportion of the alters are the same gender as ego?). We can also ask if the alters are themselves diverse or homogenous (Marsden, 1988). For example, what proportion of the alters are female? What proportion of the alters (or alter-alter dyads) are the same gender? Similarly, we can ask how ego network composition is associated with different outcomes, for example, the strength of an ethnic identity (Lubbers, Molina, & McCarty, 2007).

Ego network data also include information about the tie itself. This is summarized in row 3 of Table 10.1. The survey may ask respondents to report on the frequency of contact between themselves and the alter (daily, weekly, monthly, etc.), as well as the closeness and duration of the relationship. A survey may also capture whether the named alter is kin (mother, father, brother, etc.). A detailed survey could even ask where and under what circumstances the two met. Such information is useful in fleshing out the picture of an individual's social network (e.g., showing proportion of kin, how often ego sees their alters, etc.).

The fourth row summarizes the information surrounding the alter–alter ties. A researcher will ask the respondent to report on the ties that exist between alters 1 and 2, 1 and 3, 2 and 3, etc. (typically limited to a subset of all alters to curtail respondent burden). The alter–alter tie data capture the local structural patterns in the ego network. Are individuals socially connected to ego (the respondent) also connected to each other? Or, is a friend of a friend (the respondent) also a friend? The simplest structural measure based on the alter–alter ties is density, showing the proportion of ties that exist relative to the total number possible, but more complicated measures are possible (see Smith, 2012). Substantively, density shows to what extent people tied to ego are also tied to each other.

Applications of Ego Network Data

Ego network data thus contain information about the number of alters, the characteristics of the alters, and the structure of the ego network. Different research traditions employ this information in very different ways, depending on the question of interest and the scale of the analysis.

Using Ego Network Properties to Predict Individual-Level Outcomes

Ego network data are commonly used to measure individual-level network properties, which are then used as predictors of various outcomes, like health, mental health, job placement, cultural consumption, and the like (House, Landis, & Umberson, 1988; Dimaggio & Louch, 1998). Scholars in this tradition are most likely to collect and analyze ego network data as part of a traditional, (often) national-level survey.

The most common usage of ego network data is still as a means of measuring social support (Luke & Harris, 2007; Umberson & Montez, 2010). Social support is then used to predict health outcomes, such as mental health, health status, recovery from illness, etc. (Cornwell & Laumann, 2015; Perry & Pescosolido, 2015). The simplest measure of social support is the degree of the respondent: for example, the number of people the respondent can rely on for aid, providing specific health information, or providing emotional support. Generally, the more people one can rely on, the better the health outcomes (although this has its limits and having too many associates may be a stressor in itself; see Falci & McNeely, 2009). It is also possible to use the characteristics of the alters to measure social support. Here, different alters offer different resources from which ego can potentially draw. Ego networks composed of alters with higher income (financial resources), education (informational resources), and more kin (emotional resources) may lead to better outcomes.

Ego network data are useful, in part, because of the wide range of relational information that can be collected. For example, it is clear that not all connections are "good" or act as resources for ego. Social connections can also represent risks and potential stressors, and ego network data can just as easily capture such negative influences (Portes & Landolt, 1996). For example, if the relation of interest is drug partners, then those with higher degree (i.e., more drug partners) are actually at higher risk for engaging in risky behavior than those with lower degree (Lovell, 2002; Bailey et al., 2007). Similarly, the alter characteristics can capture potential stressors and risks that ego faces. For example, the distribution of (chronic) illness among named alters can be an important predictor of one's own health status, as dealing with an ill relative or friend may be a stressor, both physically and emotionally (Moreman, 2008).

The structural characteristics of the ego network (or the alter-alter ties) can also be used to predict individual-level outcomes. The basic idea is that norms are easier to maintain in a tight-knit network, where everyone knows everyone else (as social control is easier to maintain). For example, past work by Haynie (2001) explored adolescent delinquent behavior. She predicted delinquent behavior as a function of personal network features, including both composition and structure. Delinquency increased as friends' delinquency increased, but this relationship was clearly dependent on the density of the friendships: the rate of increase in delinquency was considerably higher in ego networks with high density. In such cases, a tight-knit network made it easier to maintain social control, making friends' delinquent behavior seem normative.

Ego network data not only provide information on risks and support but also can be used to proxy the availability and diversity of information available to ego. Here the focus is on the flow of information from
p. 177 outside sources to the focal respondent. The basic notion ↳ underlying such work is a powerful but simple one: that an individual who is receiving a diverse, nonredundant set of information is in an advantageous position (Burt, 1992). They would (potentially) hear about a wider range of job opportunities, new health information, new consumption goods, etc., while, importantly, having information that others do not have (Lin, 1999; Burt, 2004).

Past work has used ego network data as a means of measuring the information flow surrounding ego. We might expect individuals who have larger ego networks to receive more diverse information, as they are pulling from more sources. For example, more highly educated people tend to have larger networks and more diverse cultural consumption patterns (Dimaggio, 1987) (although more recent work has also argued that individuals with diverse taste join many groups, thus creating larger, more diverse networks; see Lizardo, 2006).

A long literature has alternatively focused on the relationships between ego and the named alters. Here, the question is whether those with more weak connections are more likely to find better positions, have more diverse consumption patterns, etc. (Granovetter, 1974). The argument is that weak ties are more likely to bridge loosely connected parts of the network and thus provide more unique information compared to strong "redundant" connections—that is, the strength-of-weak-ties hypotheses (although note that the evidence is mixed here) (Granovetter, 1973). Work in the organizational literature has focused more on the strategic aspects of such processes, arguing that an individual who fills a "structural hole" is likely to have informational and brokerage advantages, as they connect two groups with few ties between them (Burt, 1992). Burt has a suggested a number of ego network measures to capture these features, including effective size, efficiency, and constraint (Burt, 1992). See Table 10.1.

Using Ego Network Data to Measure Social Boundaries

Ego network data can also be used to study the social boundaries that exist in a population of interest, in addition to measuring individual-level predictors. Ego network data contain information on the demographic characteristics of both the respondents and alters, for example, offering information on education, gender, age, etc. Taken together, the respondent and alter characteristics can be used to explore homophily, or the tendency for similar individuals to interact (McPherson, Smith-Lovin, & Cook, 2001; Smith et al., 2014). Aggregating over all respondents yields the frequency of social ties between different demographic groups. One can ask how many ties exist between people of the same characteristics, or in-group ties (male-male, female-female), versus ties between people of different characteristics, or out-group ties (male-female). For example, how frequently do individuals form social connections with someone of a different race, education, or gender?

Ego network data thus offer a means of measuring the salience of a demographic dimension (Laumann, 1966; Blau & Schwartz, 1984). A low-salience, or unimportant, dimension will exhibit low levels of homophily, as the social boundaries are porous and individuals are able to freely form close social ties with members of a different social group. One can then use ego network data to tease out which demographic dimensions are more salient or are more important in organizing a social system (Marsden, 1987). Importantly, this can be done at a national level (as the data are easy to collect as part of a traditional national survey). One can ask, for example, if race/ethnicity is a more important demographic divide p. 178 ↳ than religion, age, or gender (Marsden, 1988; Rosenfeld, 2008). If collected over time, it is possible to ask how the salience of different dimensions increase/decrease in light of larger macro structural shifts, such as demographic changes (e.g., the country is more racially diverse), increasing inequality, and changes to residential segregation (Smith et al., 2014). It is also possible to make cross-national comparisons, where different countries, with very different political and economic systems, may be socially divided in different ways (Kalmijn, 1994).

Ego network data also make it possible to look at the specific divides between social groups. Are college graduates particularly unlikely to make out-group ties? Similarly, are ties between college graduates and high school graduates more or less likely than ties between college graduates and PhD holders (Smith et al., 2014)? This can be represented by a frequency table, or mixing matrix, capturing the frequency of ties between each pair of categories (e.g., Merli, Moody, Mendelsohn, & Gauthier, 2015). Such analyses can shed light on which social divides are particularly salient.

Analytically, a researcher can examine the raw, or absolute, rate of contact between demographic groups, as well as the rate of contact relative to some baseline model of chance expectations, for example, based on the demographic composition of the population of interest. The raw rates show the actual level of contact between groups (Blau, Beeker, & Fitzpatrick, 1984). How likely is someone with a high school degree to know someone with a college degree? The relative rates capture the salience of the demographic dimension more directly, by asking the same question net of what we expect by chance, just based on the size of different groups (i.e., we would expect more ties between two large groups just by chance). A number of models can be employed to handle the relative-to-chance analysis: including traditional log-linear models, case-control models, and exponential random graph models (Koehly, Goodreau, & Morris, 2004; Krivitsky, Handcock, & Morris, 2011; Smith et al., 2014).

Using Ego Network Data to Improve RDS Estimation

Ego network data have also recently been applied to more technical problems, serving as a useful corrective for other sampling traditions—most notably acting as a means of adjusting RDS estimates. RDS is a widely used sampling strategy, appropriate for conditions where no clear sampling frame exists, such as with hidden, vulnerable populations like drug users and female sex workers (Salganik & Heckathorn, 2004). RDS begins with a small number of seeds (often selected based on convenience); the initial seeds then pass on a coupon inviting other drug users, for example, into the study. These invited cases are brought into the study and the process is repeated again. Typically, each respondent will recruit two or three other people into the study.

The sampling process is thus based on personal referrals, and a researcher must make strict assumptions about this recruitment process to arrive at valid estimates (e.g., proportion of female sex workers with syphilis) (McCreesh et al., 2012; Merli, Moody, Smith, Li, Weir, & Chen, 2015). For example, one must assume that respondents refer new participants into the study by making a random selection among their network alters; or one must assume that there is no preferential recruitment. When the assumptions of the estimators do not hold, the estimates may be biased and have inflated variances (Goel & Salganik, 2010; Lu, 2013). In practice, this is often the case, as participants differentially recruit certain types p. 179 ↳ of people into the study (e.g., female sex workers may be more/less likely to pass the coupon to people in the same physical venue). The problem is that such biases are difficult to adjust for, as traditional estimators only use information from the RDS chain itself.

Work by Lu (2013) suggests that traditional RDS estimators can be greatly improved by incorporating ego network information from the recruited respondents. The basic idea is to collect ego network data on the individuals recruited into the study. The alter characteristics (from the ego network survey) are then used to adjust for any violations of the assumptions, such as differential peer recruitment. This is possible as one can compare the characteristics of the recruited peers (i.e., those they passed the coupon to) to the characteristics of the full list of alters, including those that did not receive the coupon. One can then adjust for any biases in recruitment. Lu found that using the ego network data greatly improved the estimates, with lower bias and variance than estimates based only on the RDS data itself (see also Verdery et al., 2015). More generally, this hints at the potential payoff from combining ego network data with other data sources.

Using Ego Network Data to Infer Full Network Features

Ego network data can also be used to infer global network features when full network data cannot be collected (Smith, 2012, 2015). A researcher may be interested in the network features of a population, such as distance or cohesion, where it is infeasible to collect information on all actors and all ties between actors (e.g., the network may be too large or the relationship of interest too sensitive to collect directly). It may, however, be possible to collect sampled ego network data in cases where a census cannot be collected. If it is possible to make inference about the full network structure from independently sampled ego network data, then there is a potentially radical shift in the way we think about doing network analysis—as one could collect a sample (instead of a census) and still measure the network features of a given population (e.g., Handcock & Gile, 2010; Krivitsky et al., 2011). This would, for example, make it easier to move beyond small, institutionally bounded populations.

Using sampled network data makes data collection easier but also raises difficult inference problems. It is difficult to capture network-level measures like cohesion, group structure, or diffusion potential if we cannot map out the direct and indirect connections among all actors in a network. The sampled data offer only independent pieces of the network, thus offering no direct way of measuring global network features. Instead, a researcher must infer what the full network structure looks like just based on the local information found in the ego network data.

Past work has used simulation as a solution, where the goal is to generate complete networks that are consistent with the local information found in the sampled data (Morris et al., 2009; Smith, 2012). As the ego networks are drawn randomly from the population, any network consistent with the sampled information is a possible realization of the true network. The basic idea is to gather information from the ego networks, generate full networks that have those properties, and use the generated networks to answer substantive questions about the population of interest.

Much of the literature using ego network data to generate full networks has focused on sexual networks p. 180 (Morris et al., 2009; Merli, Moody, Mendelsohn, & Gauthier, 2015). Sexual ↳ networks are a prime candidate for such analyses: it is difficult to collect full network data on sexual relations,[1] but it is necessary to know something about the global network features to characterize the epidemic risk in a population. A typical study will take information on the degree distribution, differential degree, and homophily (i.e., sexual mixing between groups) and generate a full network that is consistent with what is observed in the actual ego network data. This is generally accomplished using exponential random graph models (ERGMs) or similar approaches. Once the network is generated, it is possible to explore the potential for infectious diseases such as HIV or hepatitis C virus to spread through the network (Morris & Kretzschma, 2000).

Recent work has also used ego network data and simulation to explore the properties of spatially embedded networks. An unbounded network (i.e., not bounded by school walls or organizational affiliation) can range over a wide geographic area while still having properties that are shaped by geography; for example, interaction decreases as physical distance between two individuals increases (Spiro, Almquist, & Butts, 2016). Ego network data are useful for capturing the spatial network properties of unbounded populations. A researcher could randomly sample respondents, collecting information on the neighborhood, city, state, etc., of the respondents and alters. If physical location cannot be collected, it is possible to ask how far (or how long) each alter is from ego. This information could then be used to inform the simulation of the full network, simulating ties based on the observed physical distance between ego-alter pairs in the data (e.g., Butts et al., 2012; Merli, Moody, Smith, Li, Weir, & Chen, 2015).

It is important to note that studies in the sexual network tradition (as well as the spatial tradition) typically do not incorporate the alter-alter ties into the analysis: the alters are unlikely to have sexual relations with each other and it is difficult to ask about the sexual partners of the alters. A number of recent studies have developed models that make explicit use of the alter-alter ties (Smith, 2012, 2015). This will be particularly important in settings where one's associates are likely to know each other, the case for most (nonsexual) network ties.

The goal of such work is to incorporate the structure of the ego networks into a simulation that infers global network features from sampled data. The approach of Smith (2012), for example, offers a unique means of measuring (local) structural features from the alter-alter tie data and then generating networks based on those local features. Specifically, the alter-alter tie data are used to construct a distribution of ego network configurations. An example ego network configuration distribution is plotted in Figure 10.2 (limited to configurations with four or fewer alters for space considerations). The x-axis plots the different configurations that ego can fall into, based on the size of the ego network and the pattern of ties between alters. The y-axis captures the proportion of egos in a hypothetical sample that fall into each configuration.

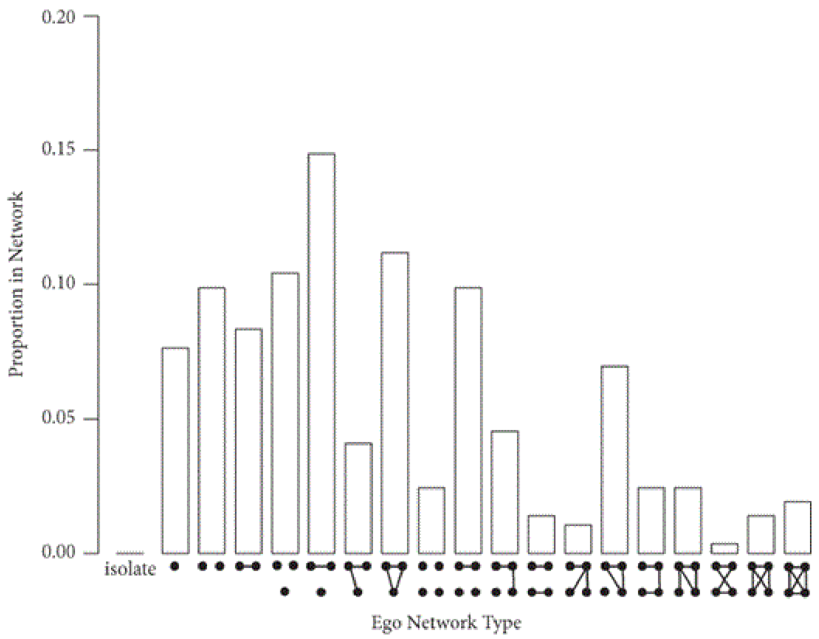

Figure 10.2 Example ego network configuration distributions.

Notes: This figure is based on a hypothetical ego network configuration distribution.Ego is not included in the ego network types. Only ego network types of size four or less are included to make the figure legible.

The basic idea is then to look for full networks with the same distribution of ego network configurations as the sampled data (the simulation is also conditioned on the degree distribution, differential degree, and homophily found in the sample). A network consistent with this local information is likely to have similar global features as the actual (unknown) network. The simulation is heavily constrained by the empirical data, making it more likely that the generated network approximates the true network. In a test of the method, Smith (2012, 2015) found that the simulation approach offers estimates of global network features (like distance and modularity)
p. 181 that closely approximate the values from the true network. ↳ This suggests that a researcher could collect sampled ego network data and still make plausible claims about the structure of the full network.

Conclusion: Future Uses of Ego Network Data

Ego network data have a long history in the social sciences, traditionally acting as a means to measure social support and social boundaries, and more recently being employed to infer global network features from a sample, as well as acting as a corrective to other sampling techniques. This long and varied history suggests something about the utility of this seemingly "too simple" network data structure. The evolution of use also suggests that ego network data might be used in very different ways in the future. We should expect innovations in methods, approaches, and substantive concerns. Here, I highlight a number of possible avenues for future work.

First, there is an opportunity to push the ego network sampling/simulation approach much further, making it a general option for network scholars. For example, one possible application is estimating contextual network effects; contextual network effects capture the relationship of global network properties, like cohesion (i.e., percent in the largest ↳ bicomponent), on individual outcomes, like mental health (Bearman & Moody, 2004). For example, do more socially cohesive schools have lower rates of depression? The question is whether one can estimate such relationships using sampled data rather than having to collect census data in every context. Researchers would first collect sampled ego network data in different contexts (schools, organizations, etc.). They would then use the sampled data to infer global network structure, yielding an estimate of cohesion, for example, across contexts. Cohesion would then be used as a contextual-level predictor of mental health, suicidality, and so on (Wray, Colen, & Pescosolido, 2011). The larger goal would be to make it easier to incorporate global network structure into traditional analyses in the social sciences, with data limitations no longer acting as an obstacle. The open question, of course, is whether sampled data will actually yield empirically valid, accurate estimates of contextual network effects.

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In a similar way, there is an opportunity to combine network sampling with agent-based models, with the goal of specifying agent-based models that are more fully grounded in empirical data. Agent-based models are based on simulated worlds where actors interact based on a set of prespecified rules; these models are used to explore/test social theories under controlled conditions. Agent-based models are (generally) based on a stylized hypothetical world, where the conditions of the simulation are not informed by empirical data.[2] Purely theoretical models, while potentially useful, are also easier to dismiss, and recent work has pushed for agent-based models to be more empirically grounded (Bruch & Atwell, 2015; Hedström & Manzo, 2015). One way of satisfying this call would be to incorporate ego network data into agent-based models, most obviously in models based on network processes, such as diffusion (e.g., diffusion of culture through a network). A researcher would use sampled ego network data to generate realistic full networks, which could then be used within the agent-based model as the base substrate for the simulation. The advantage of such an approach is that the data are easy to collect but the simulations are still conditioned on a realistic network structure.

There is also room to push the modeling of ego network data. For example, Krivistky and Morris (2015) use ego network data to estimate ERGMs (where the researcher predicts the network of interest, the ties between all *i,j* pairs, as a function of different network counts). Thus, one would only need to collect sampled data to estimate these increasingly popular statistical network models. More work on the statistical properties of these models, as well as more practical applications, is needed.

Future work should also grapple more explicitly with the limitations of ego network data. For example, respondents may forget to name alters that should have been part of their ego network (Marin, 2004). This can distort the number of alters listed (degree), as well as the characteristics of the alters. More work is needed to describe the biases that exist. More work is also needed on the survey side, showing what survey protocols can limit threats to validity. For example, Marin and Hampton (2007) suggest using multiple name generators, as certain questions may prime different associations for the respondent. It would also be useful to explore the benefits of using specific exchanges (borrow money, get a ride from, etc.) instead of more general name generators (who do you discuss important matters with?). Similarly, there is much work to be done on fully incorporating multiple relations into the analysis of ego network data. Ego network data based on multiple relations has many advantages: methodologically, the data are less prone to recall bias (as the respondent has more than one chance to name an alter); substantively, multiple relational data open up new avenues for analyses—for example, making it possible to enumerate social roles. Finally, future work should continue to explore the costs and benefits of using different collection technologies when eliciting contacts.

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Finally, there is much work to be done on combining ego network data with other data sources. We saw earlier how ego network data can improve RDS estimates, but there are a number of other ways that ego network data can be paired with additional data sources to create a unique, robust set of information. For example, it is possible to supplement ego network data with a very limited snowball sample. With a snowball sample, a researcher will contact a random subsample of all named alters, interviewing those selected alters in a separate interview. Even going out one step into the network offers a great deal of supplemental information. By interviewing the alters, one receives firsthand reports on their characteristics; this serves as a robustness check on the original reports from ego on the alters. The alter interviews also make it possible to measure asymmetry in relationships, as one can see if the named alter reports the tie as being as equally strong as the respondent did. Another possibility is to supplement the ego network data by matching the named alters to the interviewed respondents, using nicknames and descriptions (but not actual ids) as a means of identifying the alter in the list of respondents (Mouw et al., 2014). This matching process would make it possible to fill in some of the ties between the isolated ego networks.

Similarly, ego network data can be combined with automated data sources, like cell phone records and social media websites. For example, recent studies have used Bluetooth signals on cell phones to generate colocation networks (i.e., are person *i* and person *j* in the same location at the same time?) (Oliver et al., 2015). Such data have the potential to reveal the social network structure on the population of interest. Unfortunately, unless the researcher can give cell phones to everyone in the population of interest, the data only reveal a subgraph sample on the network (i.e., only showing the ties between the sampled respondents). The problem with this approach is the likely sparseness of the data. Any two individuals in the network are unlikely to interact. A subgraph sample on a large network may yield very few ties between the sampled respondents unless the sample is quite large or the network is very dense. A subgraph sample with few ties indicates that the network is not very dense but is otherwise uninformative. A researcher can add to the cell phone (colocation) data by asking the respondents to directly describe their social contacts or elicit their ego network. Such data can serve as a means of fleshing out whatever network structure emerges from the colocation data.

In short, ego network data are easy to collect and useful for a wide variety of substantive and methodological problems. Given this flexibility and ease of use, there is every reason to believe that ego network data will continue to be a useful option for network scholars.

Notes

1. This is the case for a number of reasons: first, respondents may be unwilling to name their sexual partners; second, IRB approval may be difficult to acquire; third, the network is unlikely to be bounded; and fourth, no clear list of possible partners may exist.

2. There is already a large literature employing empirical data to inform simulations of disease spread (see earlier), but the literature on agent-based modeling and nonhealth outcomes, such as culture, inequality, neighborhood mobility, etc., has been much slower to incorporate empirical data into its simulations.

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References

Bailey, S. L., Ouellet, L. J. Mackesy-Amiti, M. E., Golub, E. T., Hagan, H., Hudson, S. M., … DUIT Study Team. (2007). Perceived risk, peer influences, and injection partner type predict receptive syringe sharing among young adult injection drug users in five us cities. *Drug and Alcohol Dependence*, *91*, S18–S29.
Google Scholar WorldCat

Bearman, P. S., & Moody, J. (2004). Suicide and friendships among American adolescents. *American Journal of Public Health*, *94*(1), 89–95.
Google Scholar WorldCat

Blau, P. M., Beeker, C., & Fitzpatrick, K. M. (1984). Intersecting social affiliations and intermarriage. *Social Forces*, *62*(3), 585–606.
Google Scholar WorldCat

Blau, P. M., & Schwartz, J. E. (1984). *Crosscutting social circles*. Orlando, FL: Academic Press.
Google Scholar Google Preview WorldCat COPAC

Bruch, E., & Atwell, J. (2015). Agent-based models in empirical social research. *Sociological Methods & Research*, *44*(2), 186–221. doi:10.1177/0049124113506405
Google Scholar WorldCat

Burt, R. S. (1984). Network items and the General Social Survey. *Social Networks*, *6*(4), 293–339.
Google Scholar WorldCat

Burt, R. S. (1992). *Structural holes: The social structure of competition*. Cambridge, MA: Harvard University Press.
Google Scholar Google Preview WorldCat COPAC

Burt, R. S. (2004). Structural holes and good ideas. *American Journal of Sociology*, *110*(2), 349–399. doi:10.1086/421787
Google Scholar WorldCat

Butts, C. T., Acton, R. M., Hipp, J. R., & Nagle, N. N. (2012). Geographical variability and network structure. *Social Networks*, *34*(1), 82–100.
Google Scholar WorldCat

Cornwell, B., & Laumann, E. O. (2015). The health benefits of network growth: New evidence from a national survey of older adults. *Social Science & Medicine*, *125*, 94–106.
Google Scholar WorldCat

Cornwell, B., Laumann, E. O., & Schumm, P. L. (2008). The social connectedness of older adults: A national profile. *American Sociological Review*, *73*(2), 185–203.
Google Scholar WorldCat

Cornwell, B., Schumm, L. P., Laumann, E. O., & Graber, J. (2009). Social networks in the Nshap study: Rationale, measurement, and preliminary findings. *Journals of Gerontology Series B: Psychological Sciences and Social Sciences*, *64*(Suppl. 1), i47–i55.
Google Scholar WorldCat

DiMaggio, P. (1987). Classification in art. *American Sociological Review*, *52*, 440–455.
Google Scholar WorldCat

DiMaggio, P., & Louch, H. (1998). Socially embedded consumer transactions: For what kinds of purchases do people most often use networks. *American Sociological Review*, *63*, 619–637.
Google Scholar WorldCat

Erbring, L., & Young, A. A. (1979). Individuals and social structure: Contextual effects as endogenous feedback. *Sociological*

Methods and Research 7: 396–430.
Google Scholar WorldCat

Falci, C., & McNeely, C. (2009). Too many friends: Social integration, network cohesion and adolescent depressive symptoms. *Social Forces*, *87*(4), 2031–2061.
Google Scholar WorldCat

Fischer, C. S. (1982). *To dwell among friends: Personal networks in town and city.* Chicago, IL: University of Chicago Press.
Google Scholar Google Preview WorldCat COPAC

Frank, O. (1971). *Statistical inference in graphs* (Doctoral dissertation). Stockholm University Stockholm, Sweden.

Freeman, L. C. (1979). Centrality in social networks: Conceptual clarification. *Social Networks*, *1*, 215–239.
Google Scholar WorldCat

Gjoka, M., Smith, E., & Butts, C. (2014). Estimating clique composition and size distributions from sampled network data. In *2014 IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)* (pp. 837–842). Toronto, ON: IEEE.
Google Scholar WorldCat

Goel, S., & Salganik, M. J. (2010). Assessing respondent-driven sampling. *Proceedings of the National Academy of Sciences*, *107*(15), 6743–6747.
Google Scholar WorldCat

p. 185 Golder, S. A., & Macy, M. W. (2011). Diurnal and seasonal mood vary with work, sleep, and daylength across diverse cultures. *Science*, *333*(6051), 1878–1881.
Google Scholar WorldCat

Gould, R. V. (2002). The origins of status hierarchies: A formal theory and empirical test. *American Journal of Sociology*, *107*, 1143–1178.
Google Scholar WorldCat

Granovetter, M. (1974). *Getting a job: A study of contacts and careers*. Cambridge, MA: Harvard University Press.
Google Scholar Google Preview WorldCat COPAC

Granovetter, M. S. (1973). The strength of weak ties. *American Journal of Sociology*, *78*, 1360–1380.
Google Scholar WorldCat

Handcock, M. S., & Gile, K. J. (2010). Modeling social networks from sampled data. *Annals of the Applied Statistics*, *4*, 5–25.
Google Scholar WorldCat

Haynie, D. L. (2001). Delinquent peers revisited: Does network structure matter? *American Journal of Sociology*, *106*(4), 1013–1057.
Google Scholar WorldCat

Hedström, P., & Manzo, G. (2015). Recent trends in agent-based computational research a brief introduction. *Sociological Methods & Research*, *44*(2), 179–185.
Google Scholar WorldCat

House, J. S., Landis, K. R., & Umberson, D. (1988). Social relationships and health. *Science*, *241*(4865), 540–545.
Google Scholar WorldCat

Kadushin, C. (2012). *Understanding social networks: Theories, concepts, and findings*. New York, NY: Oxford University Press.
Google Scholar Google Preview WorldCat COPAC

Kalmijn, M. (1994). Assortative mating by cultural and economic occupational status. *American Journal of Sociology*, *100*(2), 422–

452.
Google Scholar WorldCat

Koehly, L., Goodreau, S. M., & Morris, M. (2004). Exponential family models for sampled and census network data. *Sociological Methodology*, *34*, 241–270.
Google Scholar WorldCat

Krackhardt, D., & Kilduff, M. (1999). Whether close or far: Social distance effects on perceived balance in friendship networks. *Journal of Personality and Social Psychology*, 76: 770–782.
Google Scholar WorldCat

Krivitsky, P. N., Handcock, M. S., & Morris, M. (2011). Adjusting for network size and composition effects in exponential-family random graph models. *Statistical Methodology*, *8*(4), 319–339. http://dx.doi.org/10.1016/j.stamet.2011.01.005
Google Scholar WorldCat

Krivitsky, P. N., & Morris, M. (2015). Inference for social network models from egocentrically-sampled data, with application to understanding persistent racial disparities in HIV Prevalence in the
U.S. University of Wollongong, National Institute for Applied Statistics Research Australia.
http://niasra.uow.edu.au/publications/UOW190187
WorldCat

Laumann, E. O. (1966). *Prestige and association in an urban community*. Indianapolis, IN: Bobbs-Merrill.
Google Scholar Google Preview WorldCat COPAC

Lin, N. (1999). Social networks and status attainment. *Annual Review of Sociology*, *25*, 468–487.
Google Scholar WorldCat

Lizardo, O. (2006). How cultural tastes shape personal networks. *American Sociological Review*, *71*, 778–807.
Google Scholar WorldCat

Lovell, A. M. (2002). Risking risk: The influence of types of capital and social networks on the injection practices of drug users. *Social Science & Medicine*, *55*(5), 803–821.
Google Scholar WorldCat

Lu, X. (2013). Linked ego networks: Improving estimate reliability and validity with respondent-driven sampling. *Social Networks*, *35*(4), 669–685.
Google Scholar WorldCat

Lubbers, M. J., Molina, J. L., & McCarty, C. (2007). Personal networks and ethnic identifications the case of migrants in Spain. *International Sociology*, *22*(6), 721–741.
Google Scholar WorldCat

Luke, D. A., & Harris, J. K. (2007). Network analysis in public health: History, methods, and applications. *Annual Review Public Health*, *28*, 69–93.
Google Scholar WorldCat

Marin, A. (2004). Are respondents more likely to list alters with certain characteristics?: Implications for name generator data. *Social Networks*, *26*(4), 289–307.
Google Scholar WorldCat

p. 186 Marin, A., & Hampton, K. N. (2007). Simplifying the personal network name generator. *Field Methods*, *19*(2), 163–193. doi:10.1177/1525822x06298588
Google Scholar WorldCat

Marsden, P. V. (1987). Core discussion networks of Americans. *American Sociological Review*, *52*, 122–131.

Google Scholar WorldCat

Marsden, P. V. (1988). Homogeneity in confiding relations. *Social Networks*, *10*(1), 57–76.
Google Scholar WorldCat

Marsden, P. V. (1990). Network data and measurement. *Annual Review of Sociology*, *16*, 435–463.
Google Scholar WorldCat

Marsden, P. V. (2011). Survey methods for network data. In Carrington, P. J., & Scott, J. (Ed.), *The Sage handbook of social network analysis* (pp. 370–88). London: SAGE Publications.
Google Scholar Google Preview WorldCat COPAC

McCreesh, N., Frost, S., Seeley, J., Katongole, J., Tarsh, M. N., Ndunguse, R., … Johnston, L. G. (2012). Evaluation of respondent-driven sampling. *Epidemiology (Cambridge, Mass.)*, *23*(1), 138.
Google Scholar WorldCat

McPherson, M., Smith-Lovin, L., & Brashears, M. E. (2006). Social isolation in America: Changes in core discussion networks over two decades. *American Sociological Review*, *71*(3), 353–375.
Google Scholar WorldCat

McPherson, M. J., Smith-Lovin, L., & Cook, J. M. (2001). Birds of a feather: Homophily in social networks. *Annual Review of Sociology*, *27*, 415–444.
Google Scholar WorldCat

Merli, M. G., Moody, J., Mendelsohn, J., & Gauthier, R. (2015). Sexual mixing in Shanghai: Are heterosexual contact patterns compatible with an HIV/AIDS epidemic? *Demography*, *52*(3), 919–942.
Google Scholar WorldCat

Merli, M. G., Moody, J., Smith, J., Li, J., Weir, S., & Chen, X. (2015). Challenges to recruiting population representative samples of female sex workers in China using respondent driven sampling. *Social Science & Medicine*, *125*, 79–93.
Google Scholar WorldCat

Moody, J., & White, D. R. (2003). Structural cohesion and embeddedness. *American Sociological Review*, *68*, 103–127.
Google Scholar WorldCat

Moremen, R. D. (2008). The downside of friendship: Sources of strain in older women`s friendships. *Journal of Women & Aging*, *20*(1–2), 169–187.
Google Scholar WorldCat

Morris, M., & Kretzschma, M. (2000). A micro-simulation study of the effect of concurrent partnerships on HIV spread in Uganda. *Mathematical Population Studies*, *8*(2), 109–133.
Google Scholar WorldCat

Morris, M., Kurth, A. E., Hamilton, D. T., Moody, J., & Wakefield, S. (2009). Concurrent partnerships and HIV prevalence disparities by race: Linking science and public health practice. *American Journal of Public Health*, *99*(6), 1023–1031.
Google Scholar WorldCat

Mouw, T., Chavez, S., Edelblute, H., & Verdery, A. (2014). Binational social networks and assimilation: A test of the importance of transnationalism. *Social Problems*, *61*(3), 329–359.
Google Scholar WorldCat

Oliver, N., Matic, A., & Frias-Martinez, E. (2015). Mobile network data for public-health: Opportunities and challenges. *Frontiers in Public Health*, *3*, 189. doi:10.3389/fpubh.2015.00189
Google Scholar WorldCat

Perry, B. L., & Pescosolido, B. A. (2015). Social network activation: The role of health discussion partners in recovery from mental illness. *Social Science & Medicine*, *125*, 116–128.
Google Scholar WorldCat

Perry, B. L., Pescosolido, B. A., and Borgatti, S. P. (2018). *Egocentric network analysis: Foundations, methods, and models*. Cambridge, UK: Cambridge University Press.
Google Scholar Google Preview WorldCat COPAC

Portes, A., & Landolt, P. (1996). The downside of social capital. *American Prospect*, *26*, 18–22.
Google Scholar WorldCat

Rosenfeld, M. J. (2008). Racial, educational and religious endogamy in the United States: A comparative historical perspective. *Social Forces*, *87*(1), 1–31. doi:10.1353/sof.0.0077
Google Scholar WorldCat

Salganik, M. J., & Heckathorn, D. D. (2004). Sampling and estimation in hidden populations using respondent-driven sampling. *Sociological Methodology*, *34*(1), 193–240.
Google Scholar WorldCat

Smith, J. A. (2012). Macrostructure from microstructure: Generating whole systems from ego networks. *Sociological Methodology*, *42*(1), 155–205. doi:10.1177/0081175012455628
Google Scholar WorldCat

p. 187 Smith, J. A. (2015). Global network inference from ego network samples: Testing a simulation approach. *Journal of Mathematical Sociology*, *39*(2), 125–162. doi:10.1080/0022250X.2014.994621
Google Scholar WorldCat

Smith, J. A., & Faris, R. (2015). Movement without mobility: Adolescent status hierarchies and the contextual limits of cumulative advantage. *Social Networks*, *40*, 139–153. http://dx.doi.org/10.1016/j.socnet.2014.10.004
Google Scholar WorldCat

Smith, J. A., McPherson, M., & Smith-Lovin, L. (2014). Social distance in the United States: Sex, race, religion, age, and education homophily among confidants, 1985 to 2004. *American Sociological Review*, *79*(3), 432–456. doi:10.1177/0003122414531776
Google Scholar WorldCat

Spiro, E. S., Almquist, Z. W., & Butts, C. T. (2016). The persistence of division: Geography, institutions, and online friendship ties. *Socius: Sociological Research for a Dynamic World*, *2*. doi:10.1177/2378023116634340
Google Scholar WorldCat

Umberson, D., & Karas Montez, J. (2010). Social relationships and health a flashpoint for health policy. *Journal of Health and Social Behavior*, 51(1 Suppl.):S54–S66.
Google Scholar WorldCat

Verdery, A. M., Merli, M. G., Moody, J., Smith, J. A., & Fisher, J. C. (2015). Brief report: Respondent-driven sampling estimators under real and theoretical recruitment conditions of female sex workers in China. *Epidemiology*, *26*(5), 661–665. doi:10.1097/ede.0000000000000335
Google Scholar WorldCat

Wills, T. A., & Shinar, O. (2000). Measuring perceived and received social support. In S. Cohen, L. G. Underwood, & B. H. Gottlieb (Eds.), *Social support measurement and intervention: A guide for health and social scientists* (pp. 86–135). New York, NY: Oxford University Press.
Google Scholar Google Preview WorldCat COPAC

Wray, M., Colen, C., & Pescosolido, B. (2011). The sociology of suicide. *Annual Review of Sociology*, *37*(1), 505–528. doi:10.1146/annurev-soc-081309–150,058

Google Scholar WorldCat